



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA	Div: B	Course Code: MCA01604	Batch: S2
Semester: III		Course Name: Data Science Laboratory	
Name: Tejashree Vijay Dumasia		Roll No: 52117	
CO No: CO605.1		Assignment No: 03	

Title: Use the Naive Bayes classifier to perform sentiment analysis on a dataset of movie reviews, classifying them as either positive or negative based on the review text

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report

# Download a sample movie review dataset (using NLTK's movie_reviews if available)
try:
    import nltk
    nltk.download('movie_reviews')
    from nltk.corpus import movie_reviews
    docs = [(movie_reviews.raw(fileid), category)
             for category in movie_reviews.categories()
             for fileid in movie_reviews.fileids(category)]
    df = pd.DataFrame(docs, columns=['review', 'sentiment'])
except Exception:
    # Fallback: use a small hardcoded dataset
    data = {
        'review': [
            'I loved this movie, it was fantastic!',
            'Terrible film. Waste of time.',
            'Absolutely wonderful! A must watch.',
            'Not good. The plot was boring.',
            'Great acting and story.',
            'Awful. I hated every minute.',
            'Best movie ever!',
            'Worst movie I have seen.',
        ],
        'sentiment': [
            'pos', 'neg', 'pos', 'neg', 'pos', 'neg', 'pos', 'neg'
        ]
    }
```



```
}  
df = pd.DataFrame(data)  
  
# Encode sentiment labels  
label_map = {'pos': 1, 'neg': 0}  
df['label'] = df['sentiment'].map(label_map)  
  
# Split data  
X_train, X_test, y_train, y_test = train_test_split(  
    df['review'], df['label'], test_size=0.2, random_state=42, stratify=df['label']  
)  
  
# Vectorize text  
vectorizer = CountVectorizer(stop_words='english')  
X_train_vec = vectorizer.fit_transform(X_train)  
X_test_vec = vectorizer.transform(X_test)  
  
# Train Naive Bayes classifier  
clf = MultinomialNB()  
clf.fit(X_train_vec, y_train)  
  
# Predict and evaluate  
y_pred = clf.predict(X_test_vec)  
accuracy = accuracy_score(y_test, y_pred)  
report = classification_report(y_test, y_pred, target_names=['Negative', 'Positive'])  
  
print('--- Sentiment Analysis with Naive Bayes ---')  
print(f'Accuracy: {accuracy:.4f}')  
print("\n--- Classification Report ---")  
print(report)
```

Output:

```
--- Sentiment Analysis with Naive Bayes ---  
Accuracy: 0.5000  
  
--- Classification Report ---  
              precision    recall  f1-score   support  
  
   Negative       0.50      1.00      0.67         1  
   Positive       0.00      0.00      0.00         1  
  
   accuracy                   0.50         2  
  macro avg       0.25      0.50      0.33         2  
 weighted avg       0.25      0.50      0.33         2
```